

Adaptive Conformal Inference with explicit gradient step size for Time Series

Guillaume Principato

En collaboration avec Gilles Stoltz, Jean-Michel Poggi, Yvenn Amara-Ouali, Yannig Goude, Bachir Hamrouche



Outline

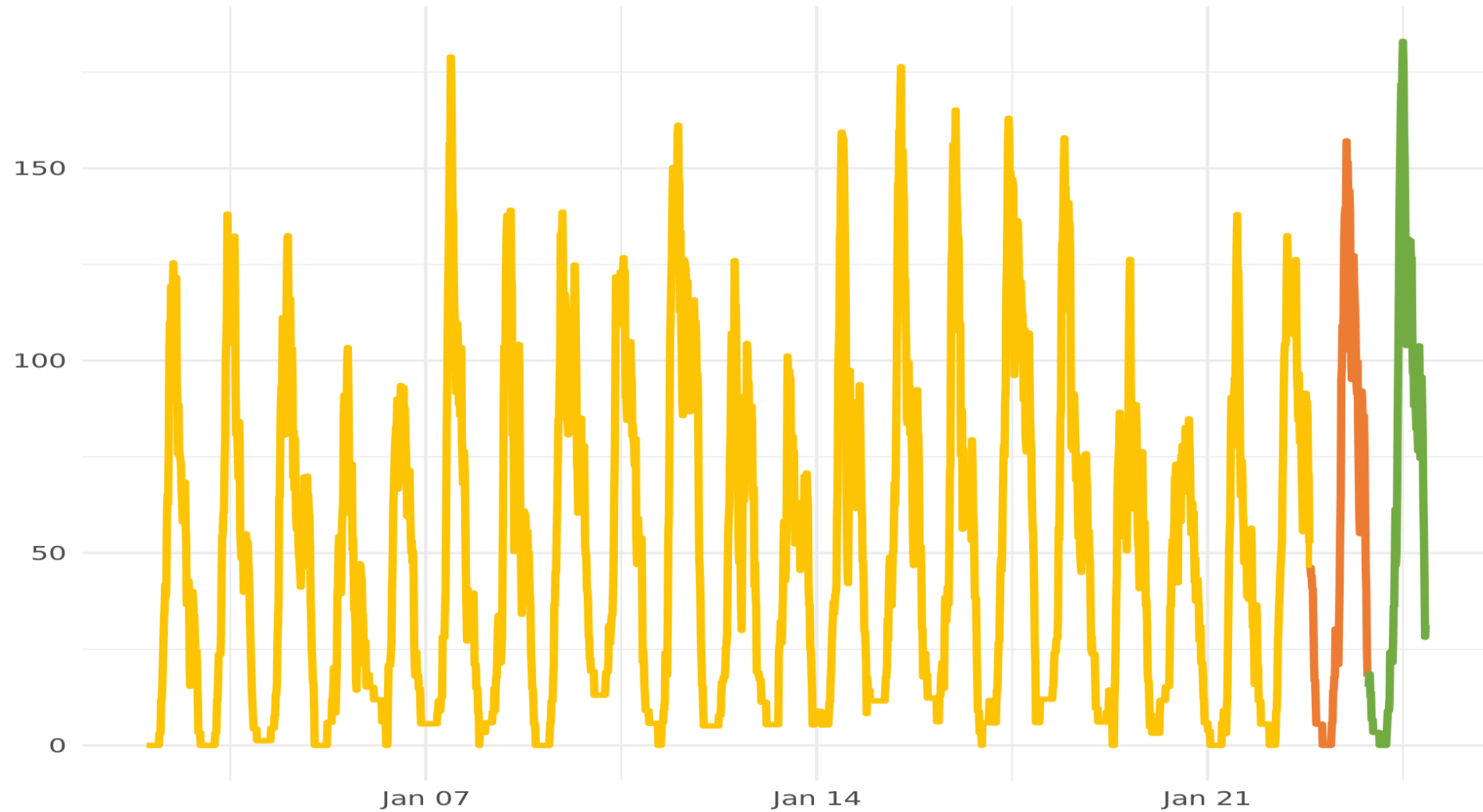
- Overview of conformal prediction for time series: Algorithm ACI.
- Tuning of ACI step size. How to get a dynamic gradient step size ?

Split Conformal Prediction (SCP)

Training

Calibration

Test

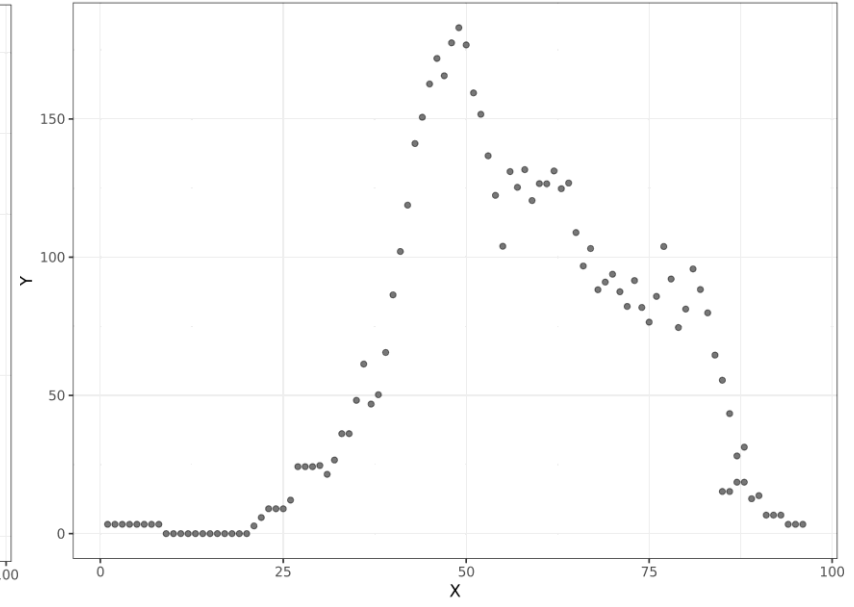
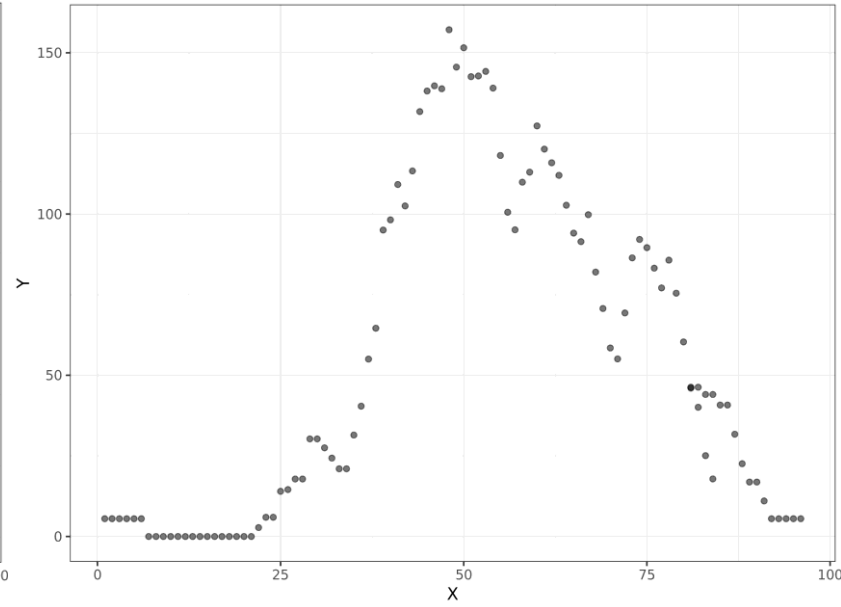
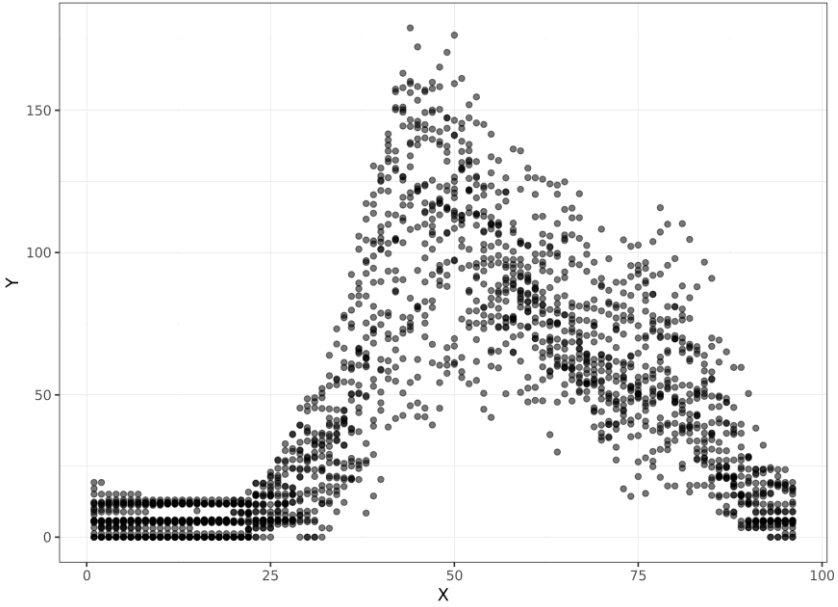


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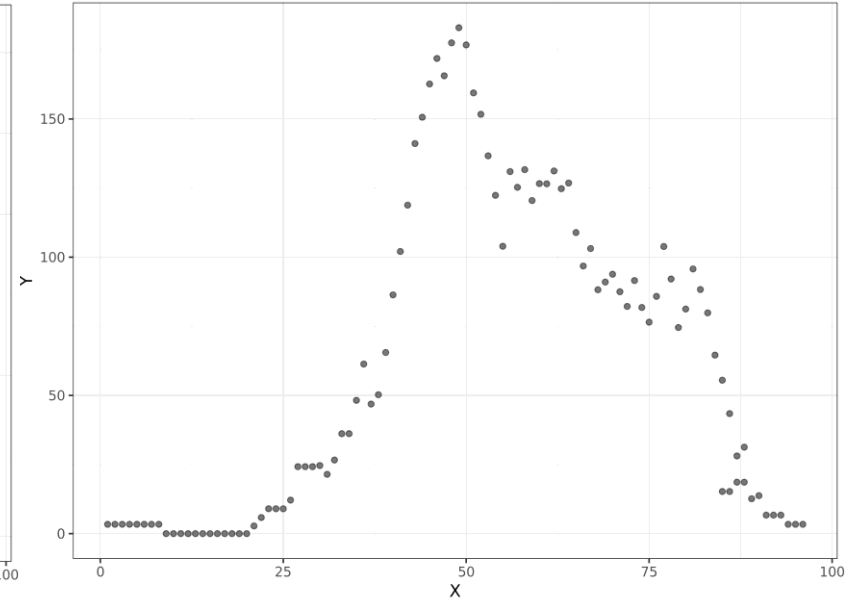
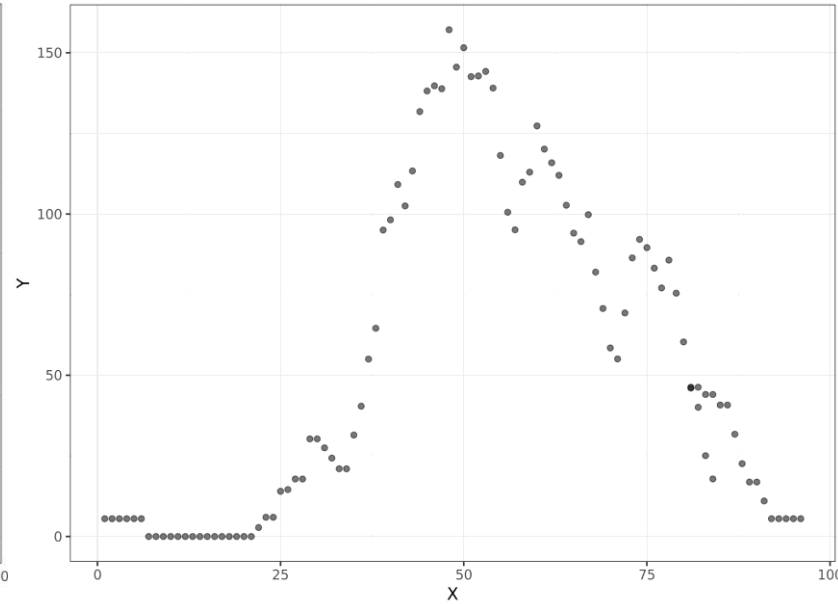
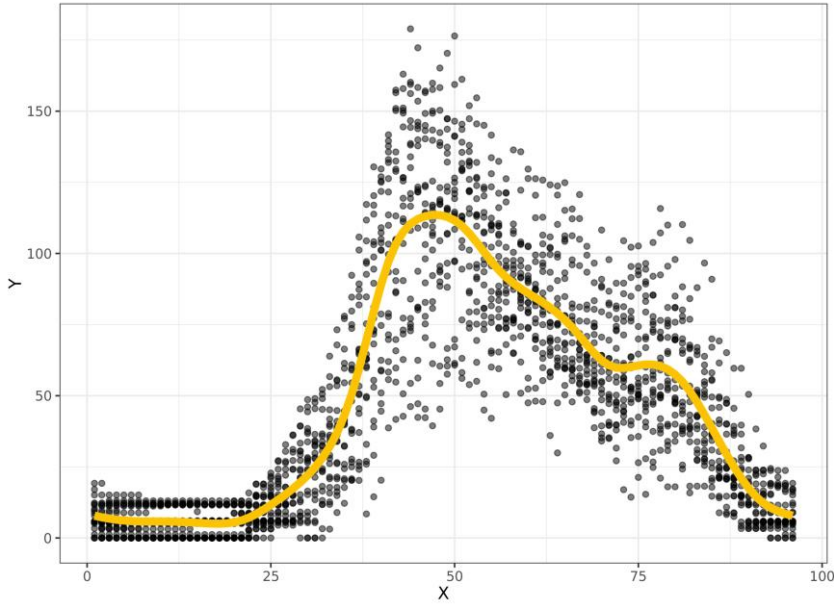


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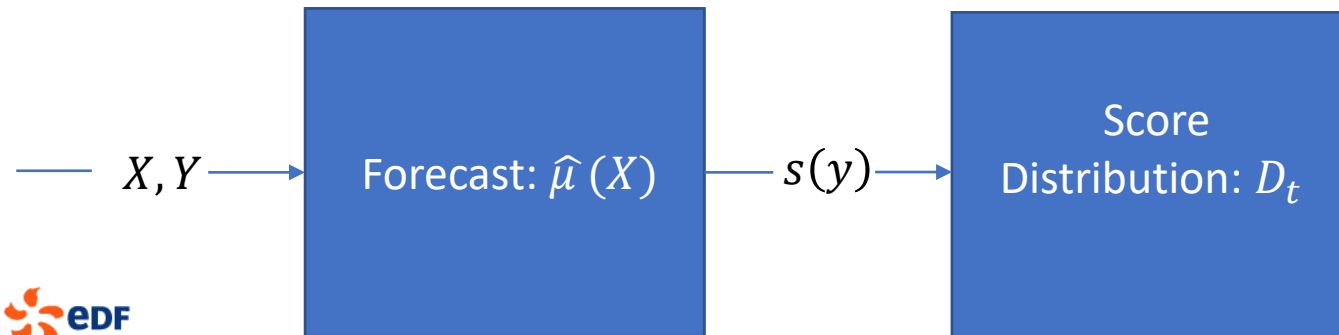
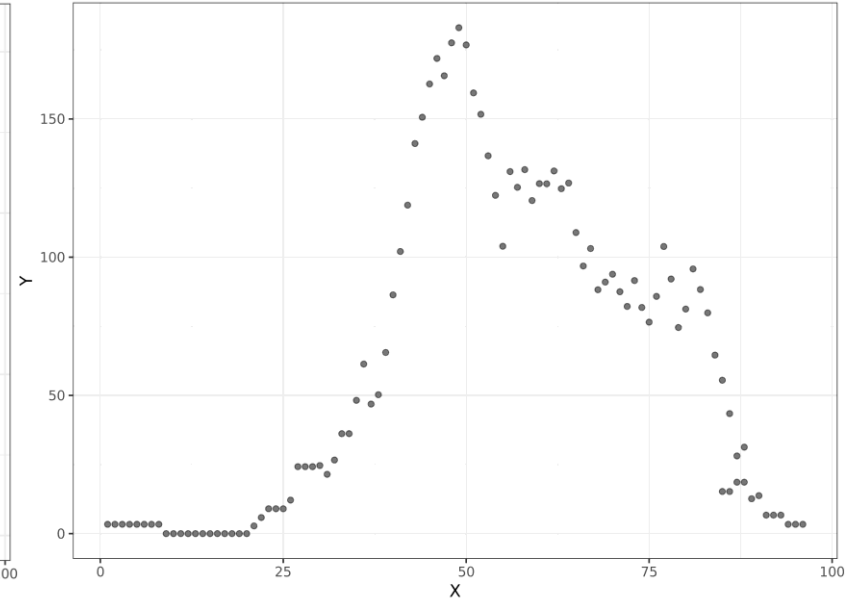
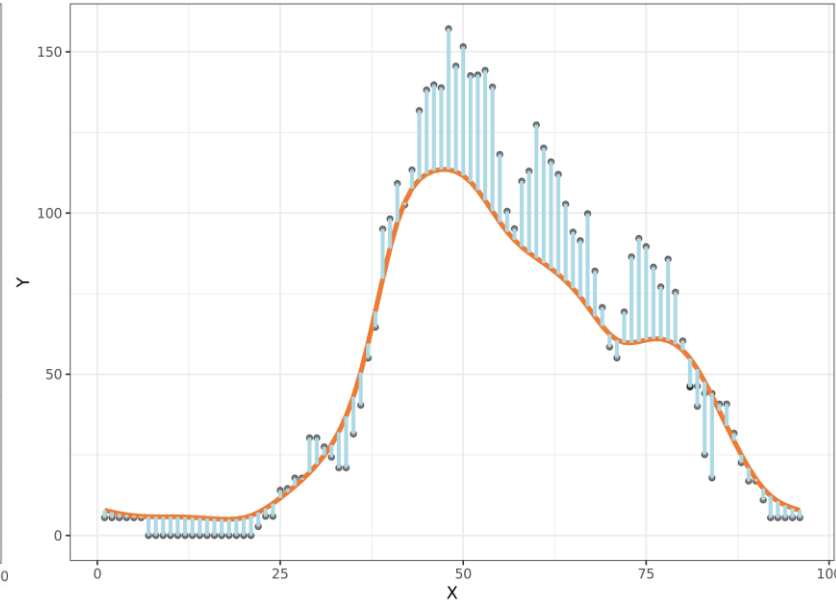
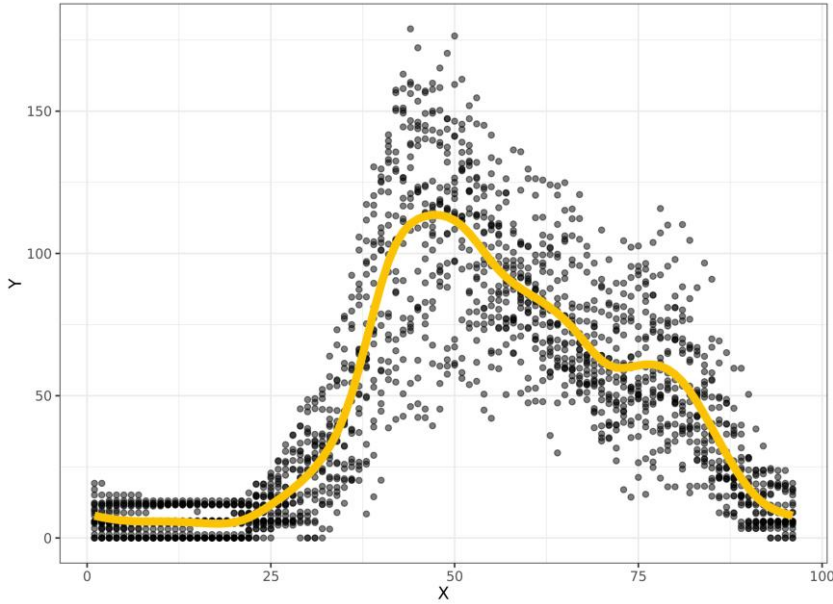
— X, Y —→ Forecast: $\hat{\mu}(X)$

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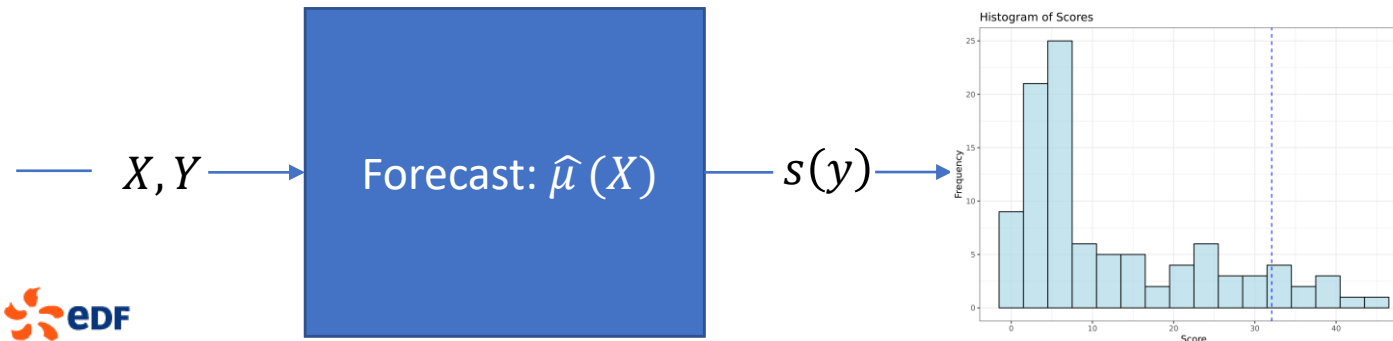
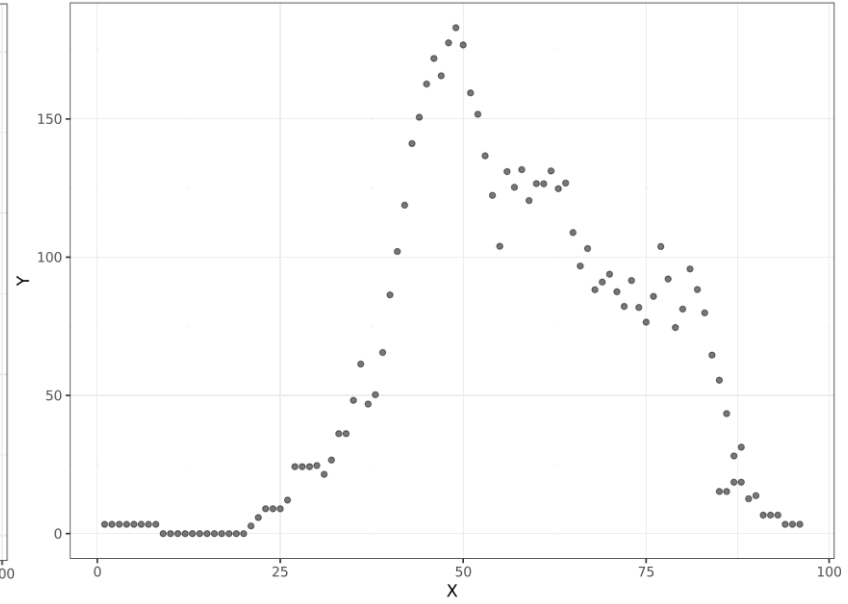
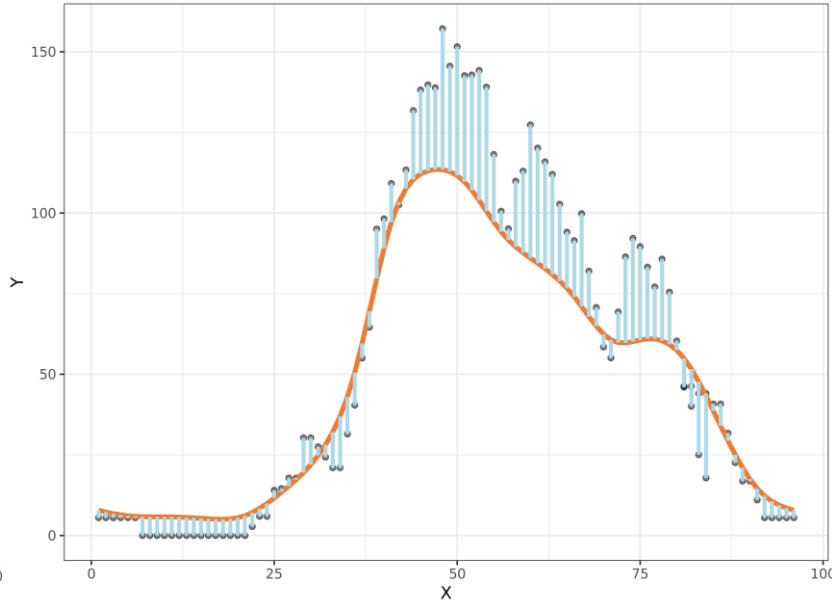
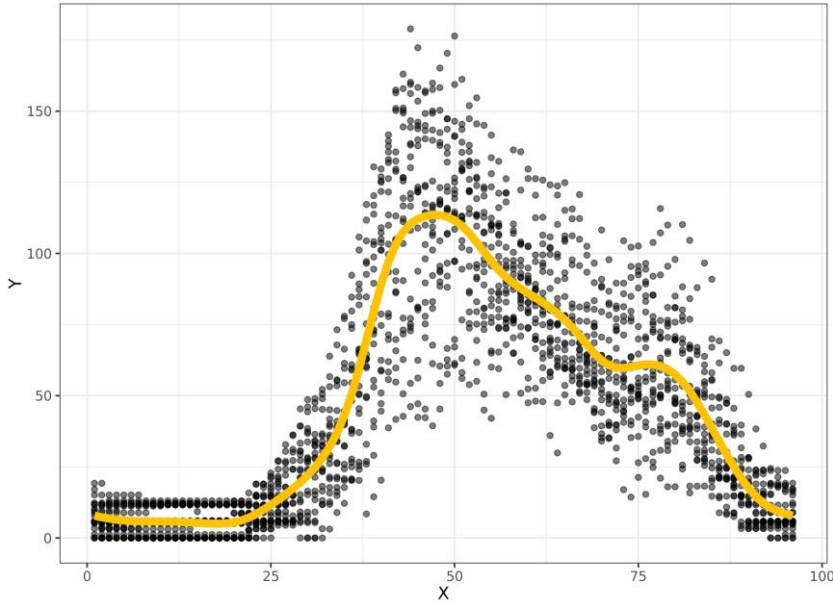


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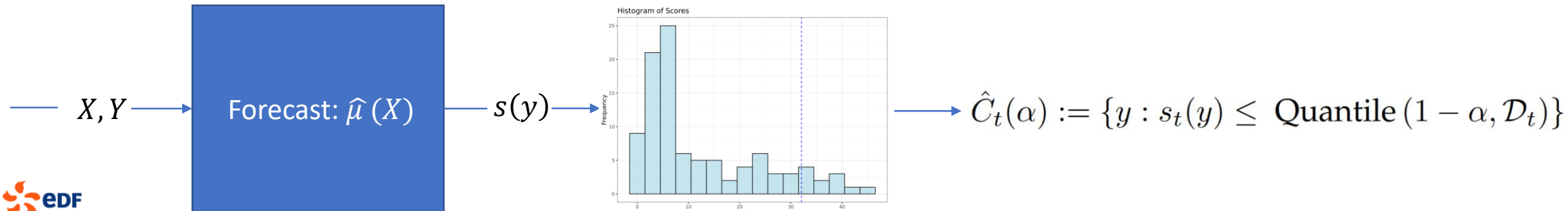
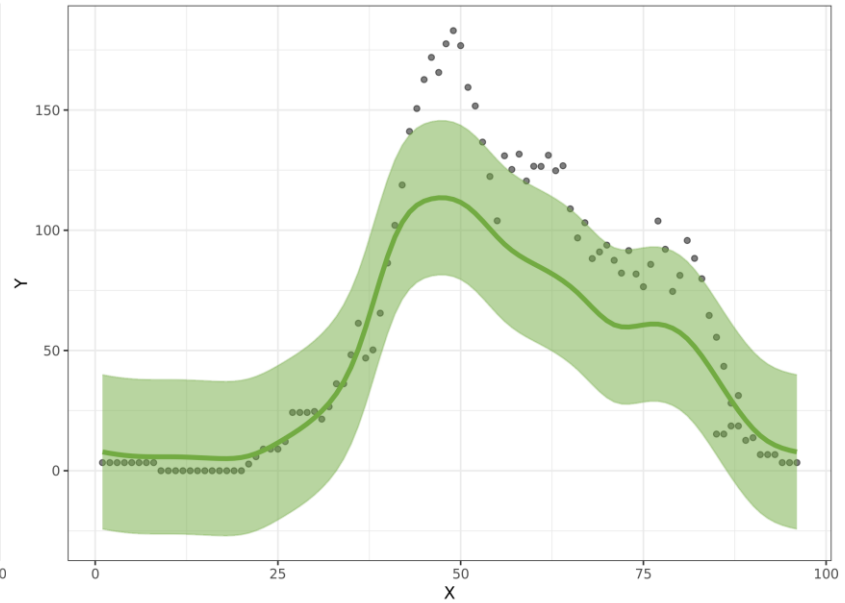
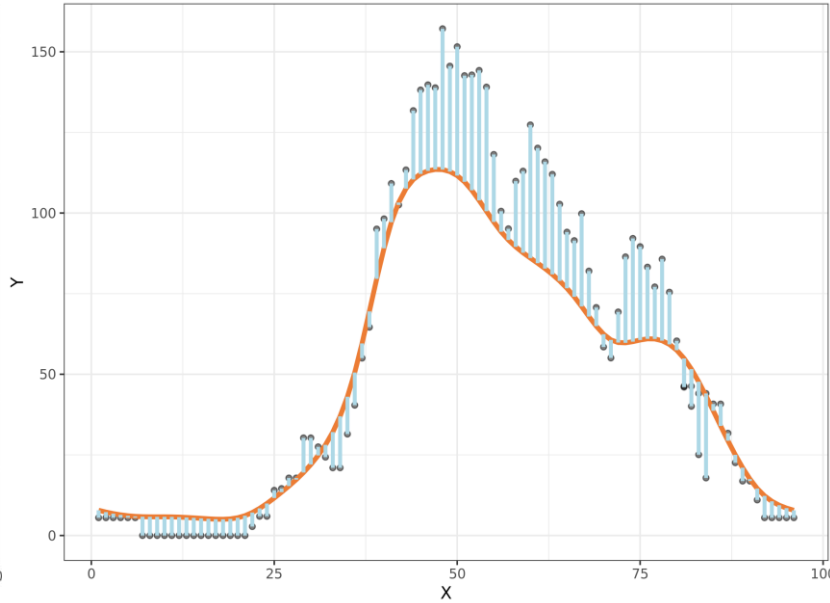
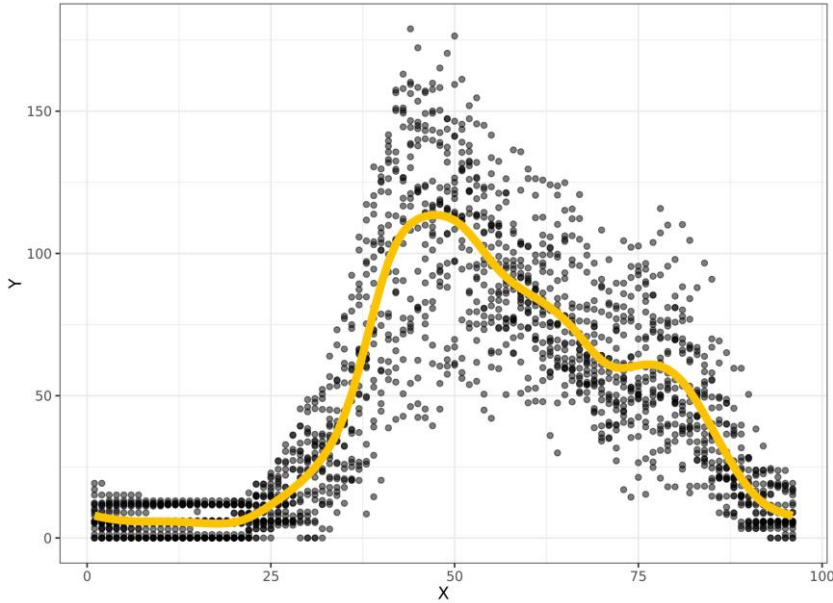


Split Conformal Prediction (SCP)

Training

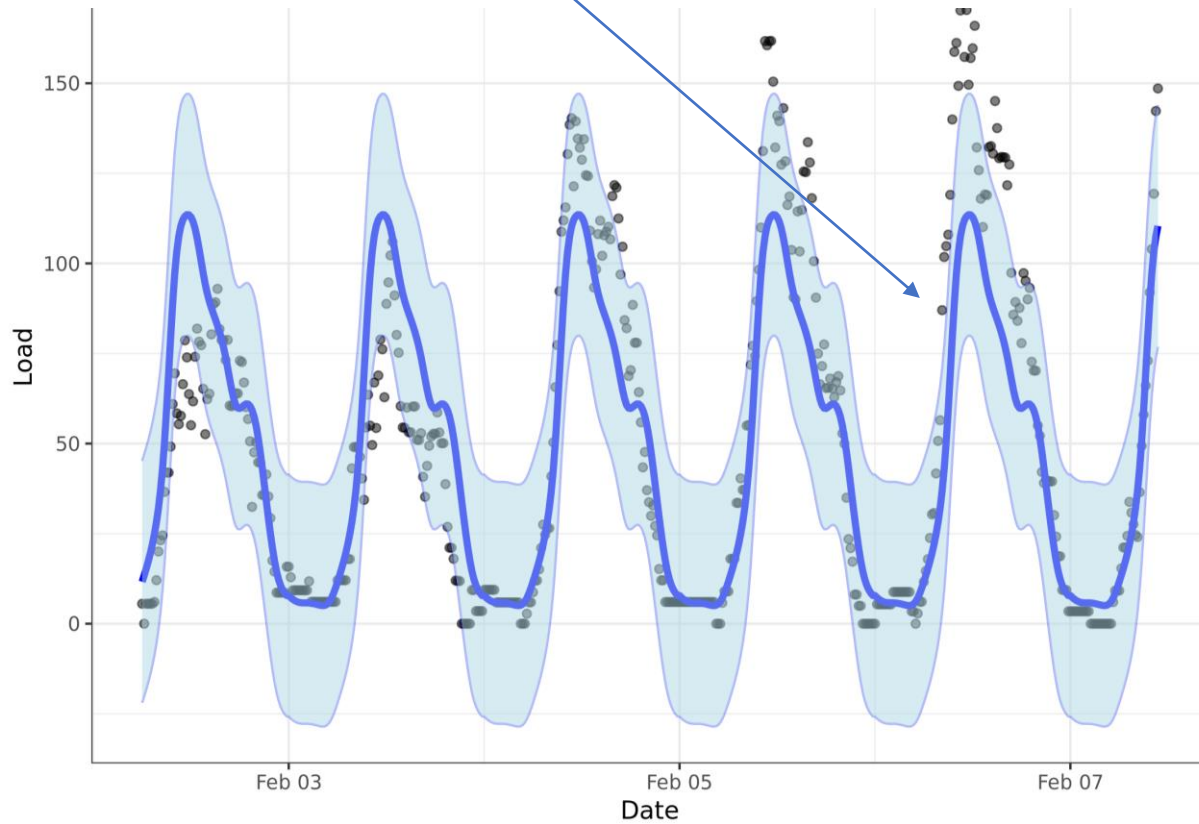
Calibration

Test

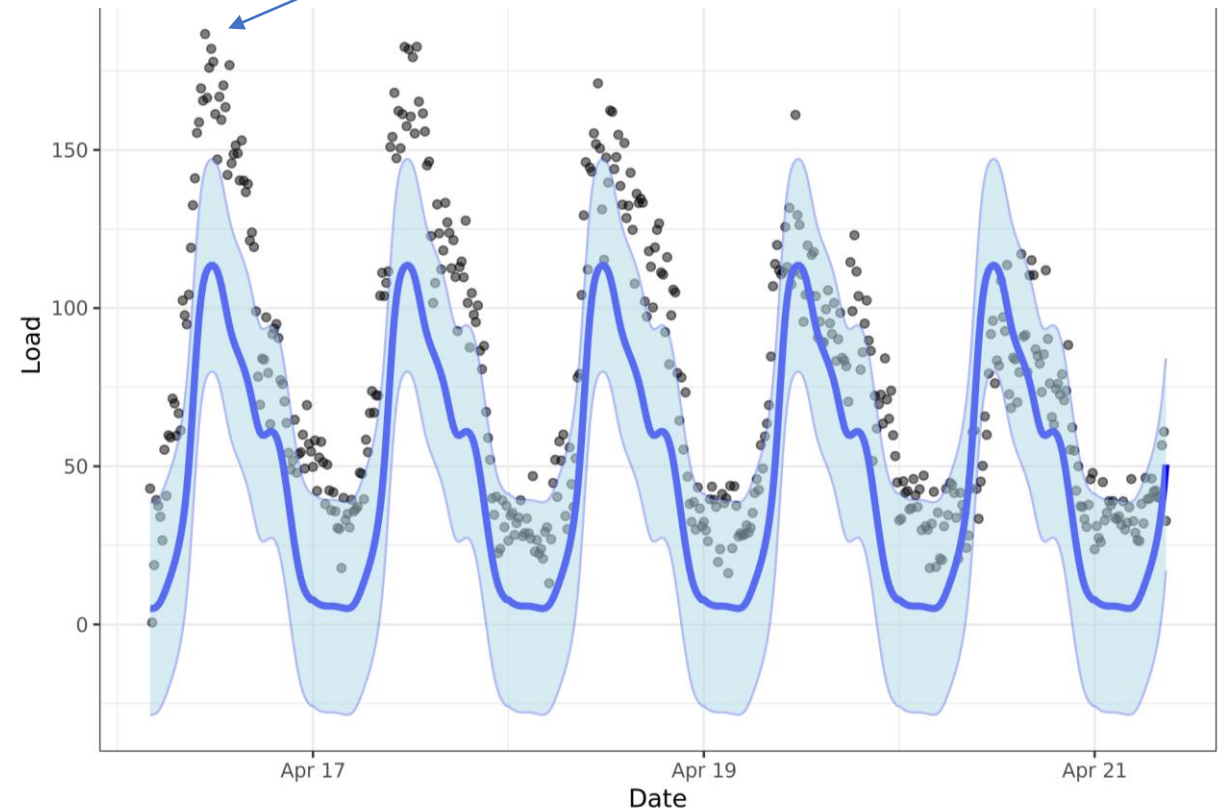


Non-exchangeable setting

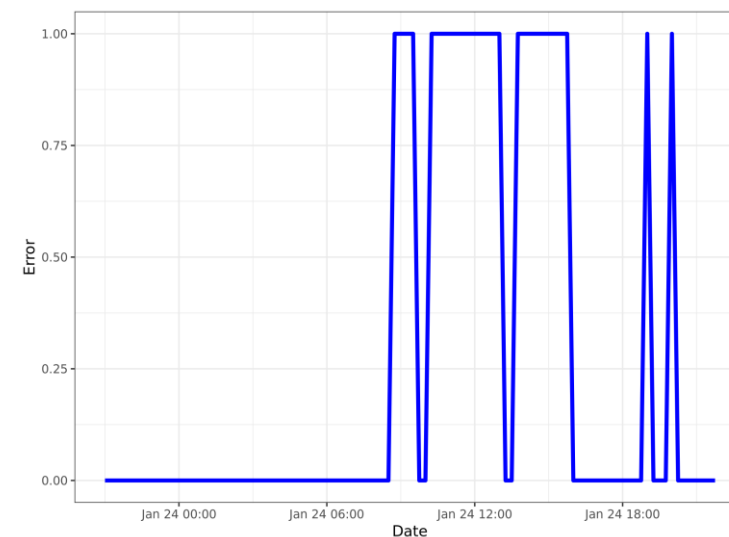
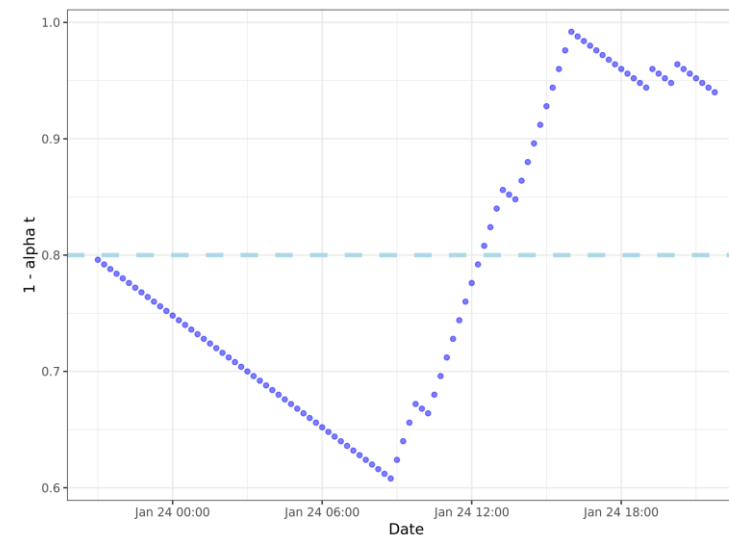
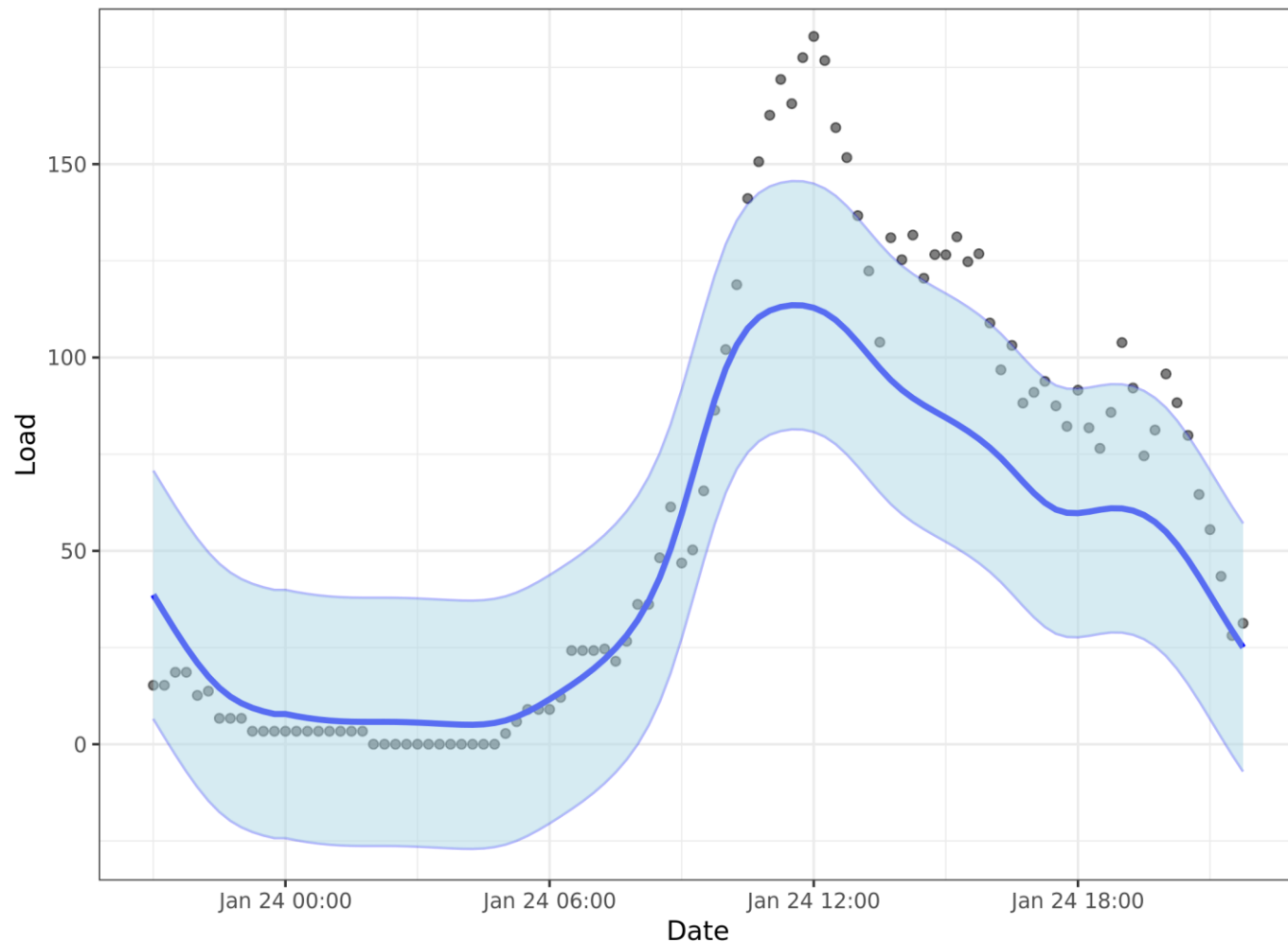
Dependency in high consumption events



Drift both in term of Mean and Deviation

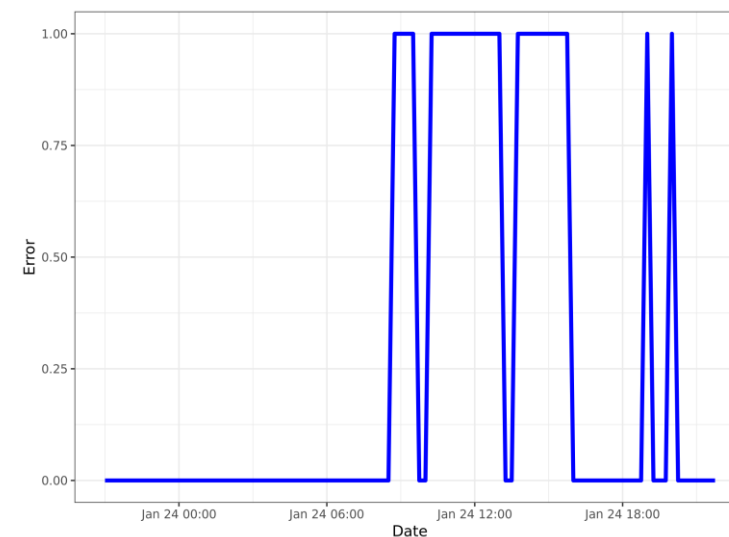
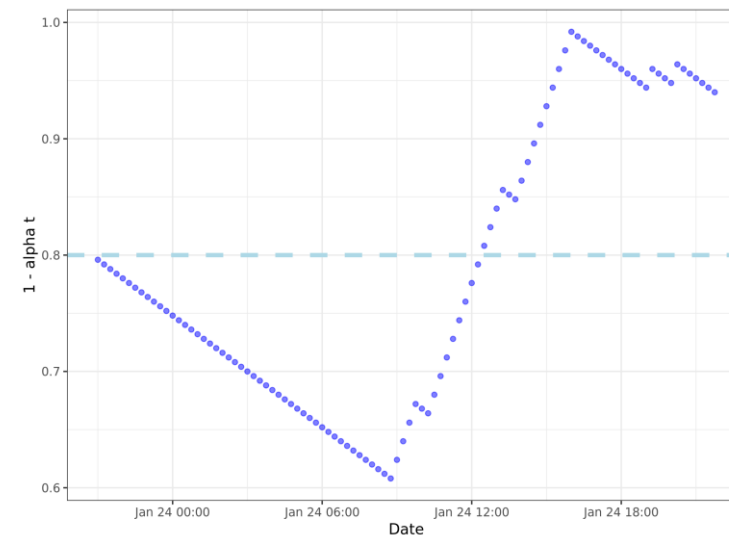
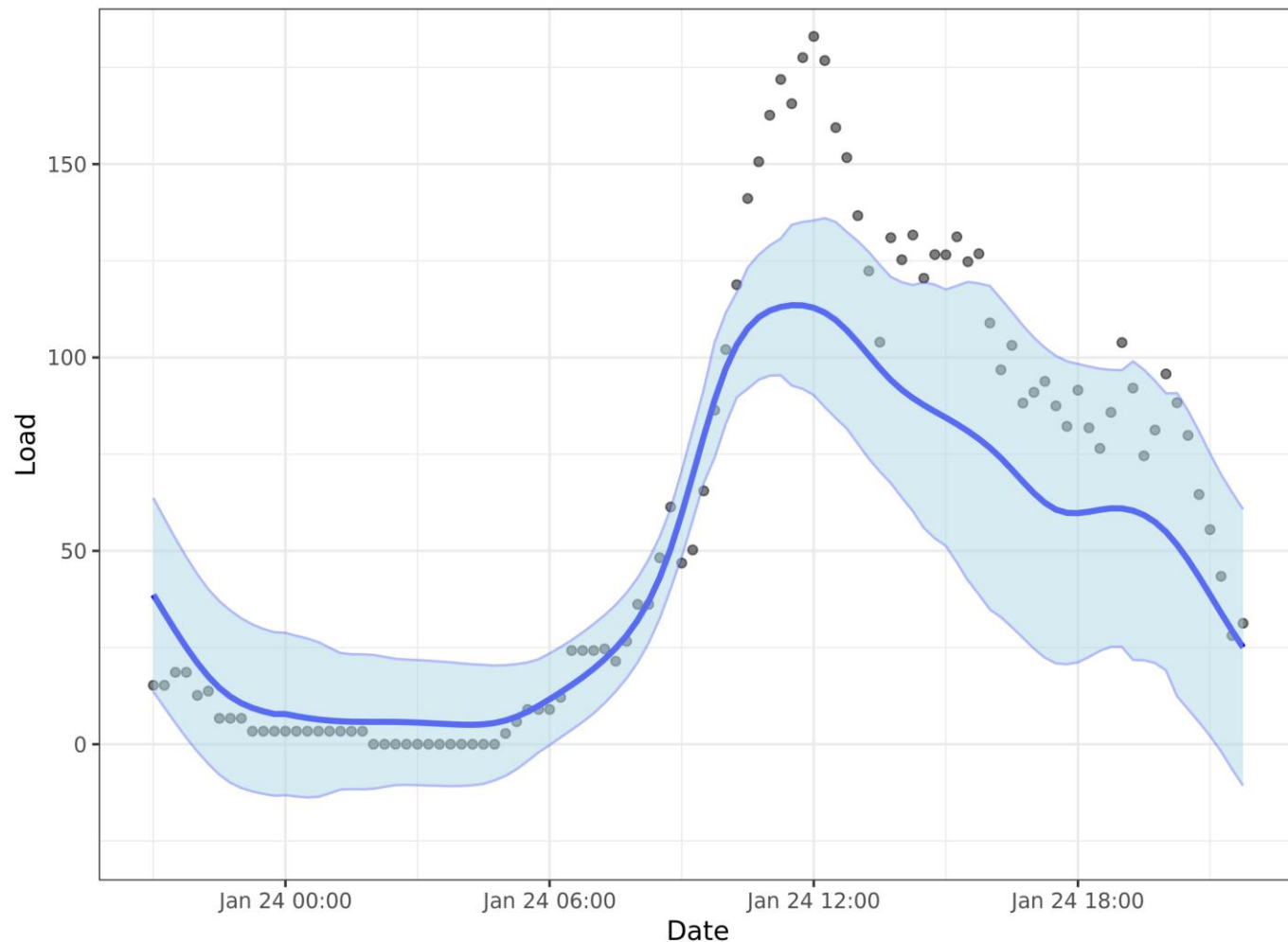


Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

How to evaluate the quality of a prediction set ?

Validity:

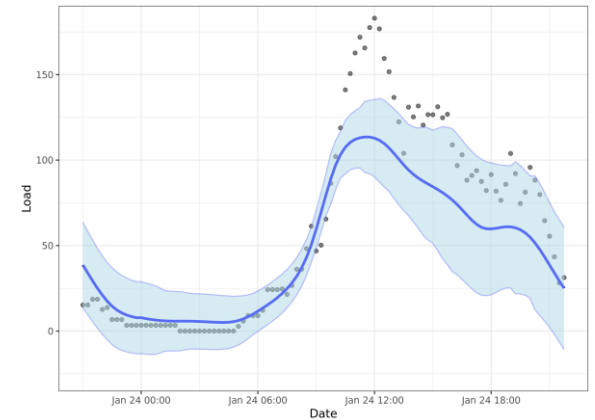
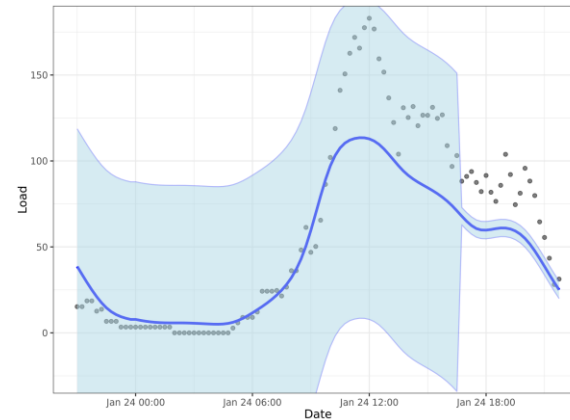
➤ Probabilistic:

$$\mathbb{P} \left(Y_{test} \in \hat{C}_{test} \right) \geq 1 - \alpha$$

➤ Asymptotic:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{\{Y_t \in \hat{C}_t\}} \stackrel{a.s.}{=} 1 - \alpha$$

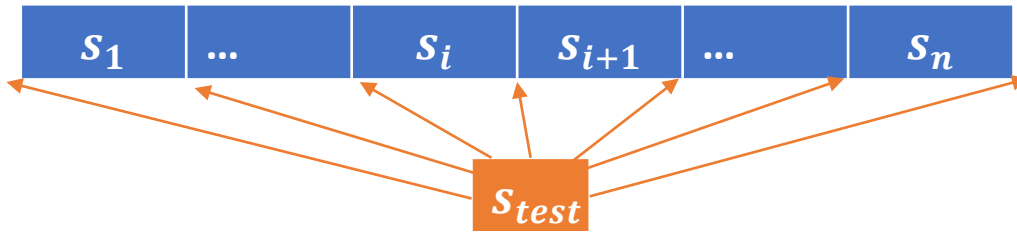
Efficiency:



- Interval width
- Regret with respect to the Pinball loss
- Tracking

Coverage guarantees and sketch of proof

Exchangeable setting



$$\mathbb{P}(s_{test} \leq s_i) = \frac{i}{n+1}$$

$$\mathbb{P}(s_{test} \leq s_{\lceil (n+1)(1-\alpha) \rceil}) \geq 1 - \alpha$$

Time Series setting (ACI)

$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t)$$

$$\frac{1}{T} \sum_{t=1}^T (\alpha - \text{err}_t) = \frac{\alpha_{T+1} - \alpha_1}{T\gamma}$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{\{Y_t \in \hat{C}_t\}} \stackrel{a.s.}{=} 1 - \alpha$$

- This proof only works with constant γ
- There is a tradeoff with the choice of γ . The higher it is the faster we get the coverage but the more instable the algorithm is.

How to choose the gradient step size ?

- We want to maintain the validity while being more efficient
- To tackle this issue, we need to use a **dynamic step size**:

¹Zaffran et al. (2022)

²Gibbs & Candès (2022)

³Bhatnagar et al. (2023)

⁴Angelopoulos et al. (2024)

This presentation (2024)

Meta algorithm to obtain γ_t

Explicit γ_t

¹Adaptive conformal predictions for time series / ²Conformal inference for online prediction with arbitrary distribution shifts /

³Improved online conformal prediction via strongly adaptive online learning / ⁴Online conformal prediction with decaying step sizes

Explicit gradient step-size

➤ Formula based on regret minimization: Orabona (2019)

$$\gamma_t = \frac{\eta}{\sqrt{\sum_{s=1}^t \|g_s\|^2}}$$
$$\alpha_{t+1} = \alpha_t + \frac{\eta}{\sqrt{\sum_{s=1}^t (\text{err}_s - \alpha)^2}} (\alpha - \text{err}_t)$$


- First studied in this context by Bhatnagar et al. (2023)
- We improve their analysis (both for Coverage and Efficiency)

Proposition 1: Asymptotic Coverage

$$\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{\{Y_t \in \hat{C}_t^\alpha\}} - (1 - \alpha) \right| = \left| \frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right| \leq O\left(\frac{1}{\sqrt{T}}\right)$$

Compare to:

- $\frac{1}{T}$ rate for a fixed step size
- $\frac{\log(T)}{T^{1/4}}$ with the same step size in Bhatnagar et al. (2023)
- Angelopoulos et al. (2024) independently found similar results with another proof

Proposition 2: Tracking

Let α_t^* such that $\mathbb{P} \left(Y_t \in \hat{C}_t(\alpha_t^*) \mid \text{past} \right) = 1 - \alpha$.

$$\sum_{t=1}^T \mathbb{E} \left[(\alpha_t - \alpha_t^*)^2 \right] \leq \sqrt{T} \cdot \text{Var}(\alpha_1^*, \dots, \alpha_T^*)$$

with $\text{Var}(\alpha_1^*, \dots, \alpha_T^*) = O \left(\frac{\sum_{t=2}^T \mathbb{E} \left[|\alpha_t^* - \alpha_{t-1}^*| \right]}{p} \right)$ 

Expected to be small
for Time Series

Compare to Gibbs & Candès (2022): $\sqrt{T} \cdot \sqrt{\text{Var}(\alpha_1^*, \dots, \alpha_T^*) + \log T}$

Use Case: Palo Alto EV Charging UQ

Yvenn Amara-Ouali presenting on the session: « Statistique appliquée à l'industrie »

Tomorrow afternoon, 16:20-16:35

